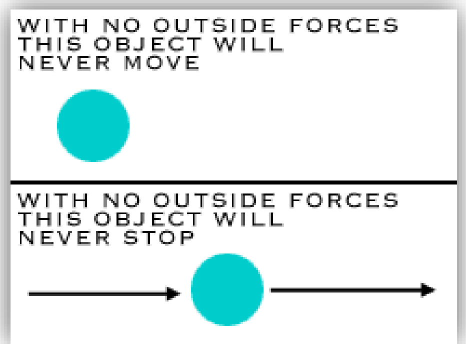
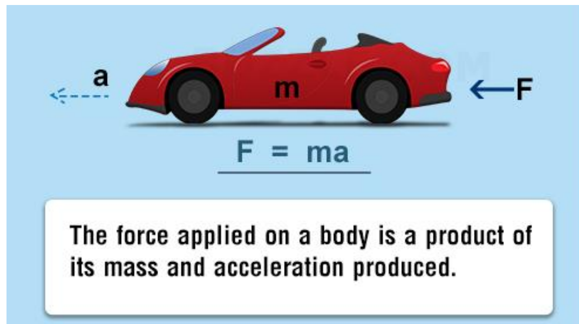
Classical Mechanics:

The development of classical mechanics is based on Newton's three laws of motions.

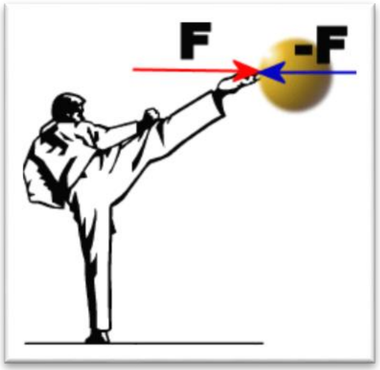


(i) Law of inertia

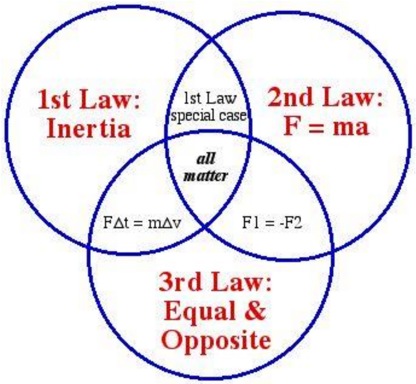


(ii) Law of force

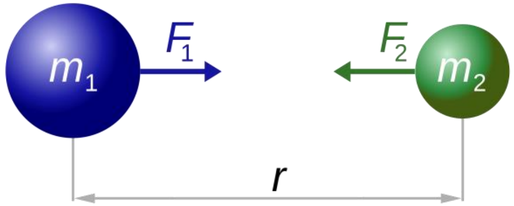
Issac Newton (1643-1727)



(iii) Law of action and reaction.



Newton's Laws



$$F_1 = F_2 = G\frac{m_1 \times m_2}{r^2}$$

Law of Universal Gravity:

What is Quantum Mechanics?

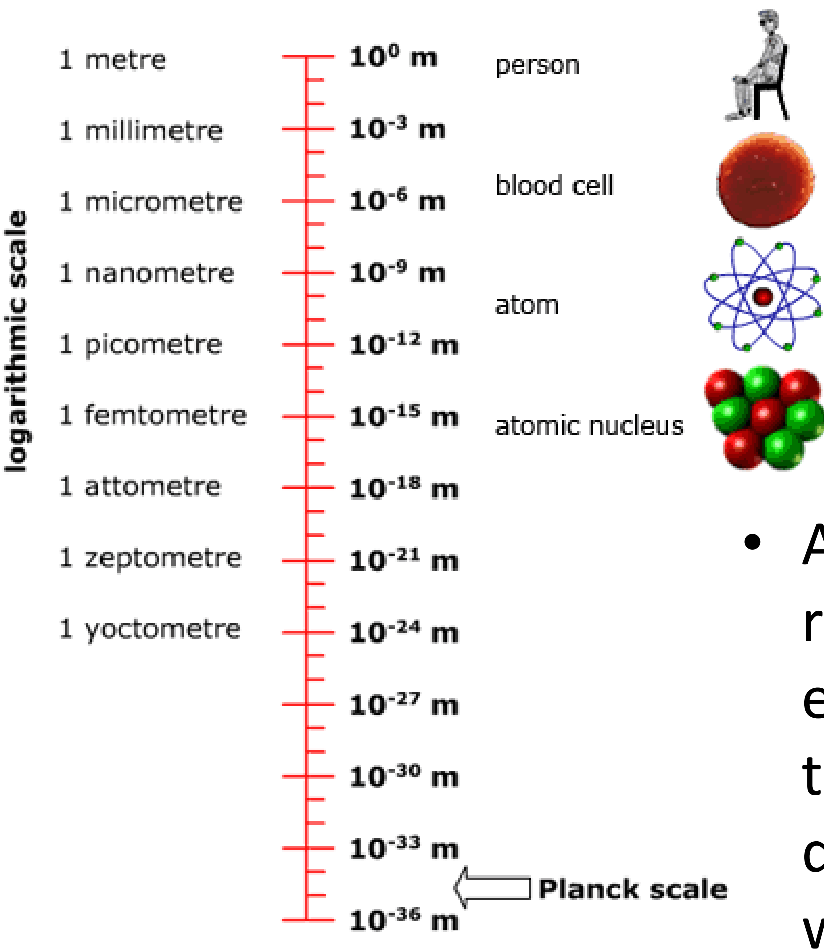
Quantum mechanics is the only theory that properly describes atomic and subatomic phenomena; it and only it explains why an atom or molecule, or even a solid body, can exist, and it allows us to determine the property of such systems.

Quantum mechanics allows to predict and understand the:

- Structure of atoms and molecules,
- Structure of bulk crystals and interfaces,
- Phase diagrams of materials and phase transitions,
- Tensile and shear strengths of materials,
- Photon spectra,
- Vibrational frequencies of condensed material,
- Specific heats of materials,
- Thermal expansion coefficients,
- Electrical and thermal conductivities,
- Surface energies, etc

Planck, Max

Nobel Prize: 1918



Light is particle!

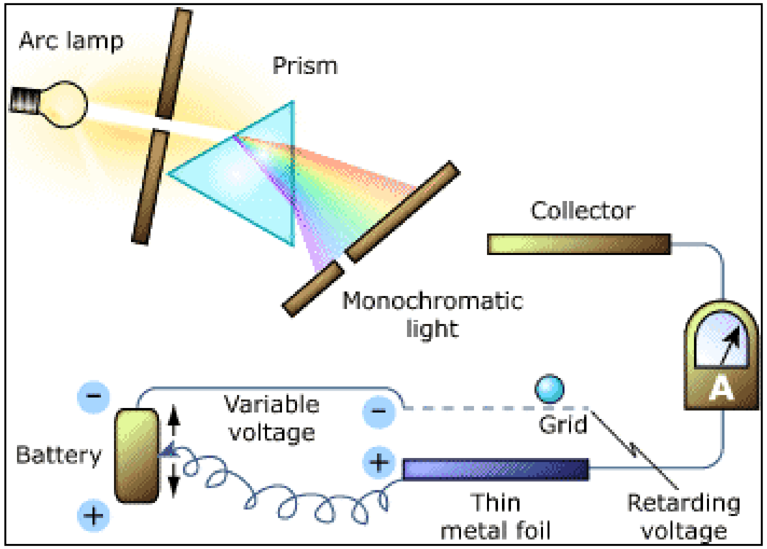
But Planck did not believe it himself: ”just a formal assumption…”

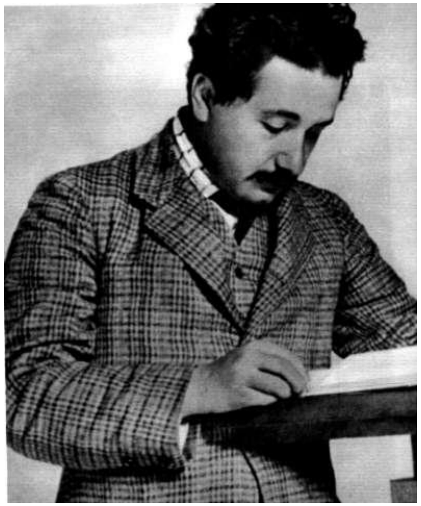
- According to Plank’s, every radiating atom in a solid emits energy only discretely in quanta, the energy of and individual quantum being equal to *hf*, ($E=hf$) where h= (6.62×10^{-34} joule sec) is Planck constant and υ is frequency

Photoelectric Effect (1887-1905)

discovered by Hertz in 1887 and explained in 1905 by Einstein.

Heinrich HERTZ
(1857-1894)





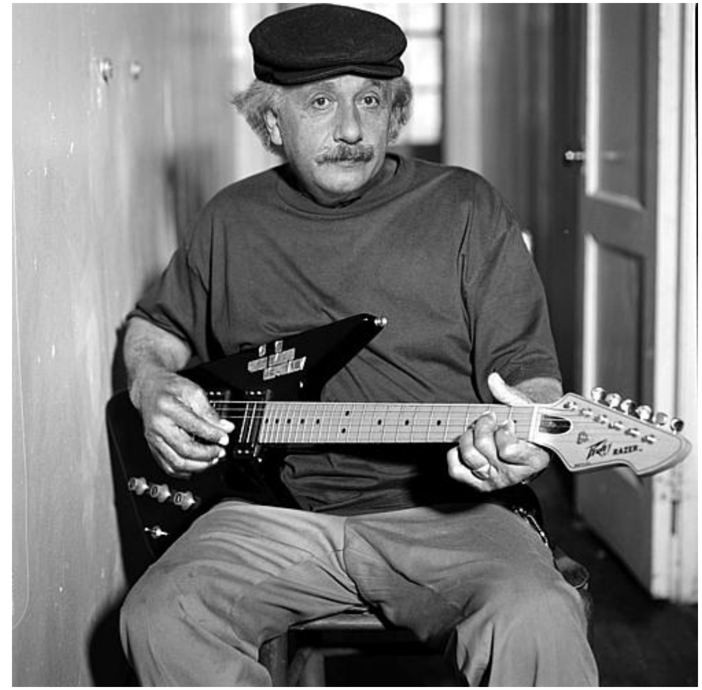
Albert EINSTEIN
(1879-1955)

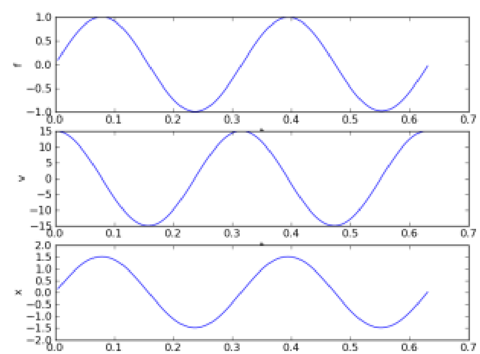
According to Einstein's, light emitted as well and absorbed discretely. He discovered in 1905. Light energy known as photon, possesses not only a definite energy *hf* but also a definite momentum equal to *hf/c* or *h/λ*.

Albert Einstein

March 1905

- He argued that light can act as though it consists of discrete, independent particles of energy, in some ways like the particles of a gas.
- For *n* number of photons, total energy *E=nhf, n=0,1,2,3,…..*
- His revolutionary proposal seemed to contradict the universally accepted theory that light consists of smoothly oscillating electromagnetic waves.





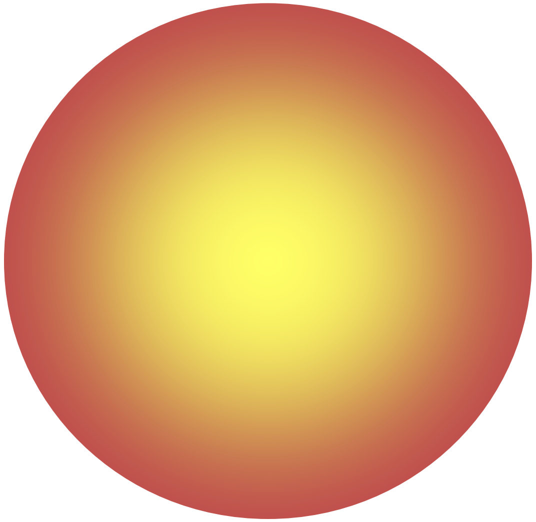
Einstein testing his theory of quanta to the resonance of his new Razor Atomic Guitar

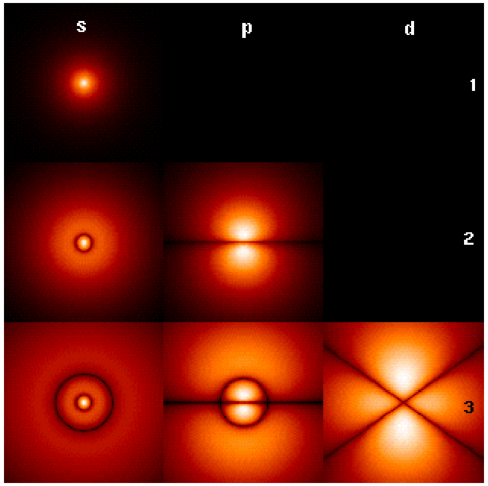
Louis de Broglie

- His 1924 doctoral thesis introduced his theory of electron waves. This included the wave-particle duality theory of matter, based on the work of Einstein and Planck.
- This research concluded in the de Broglie hypothesis stating that *any moving particle or object had an associated wave*.

Modern View

- The atom is mostly empty space
- Two regions
  - Nucleus
    - protons and neutrons
  - Electron cloud
    - region where you might find an electron





Hydrogen ~ 0.1 *nanometer* ~ $10^{-10} m$

Cricket ball;

$$m = 0.5kg \qquad v = 1ms^{-1}$$

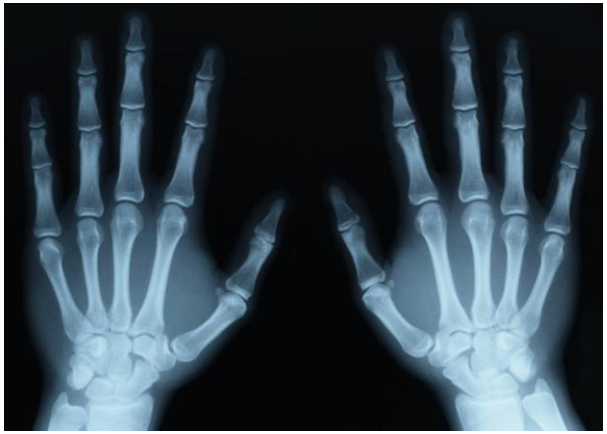
$$\therefore\ p = mv = 0.5 \times 1 = 0.5Kgms^{-1}$$

$$\lambda = \frac{h}{p}$$

$$= \frac{6.63 \times 10^{-34}}{0.5}$$

$$= 1.3 \times 10^{-33} m \ \textit{(very small wavelength)}$$

Electron;



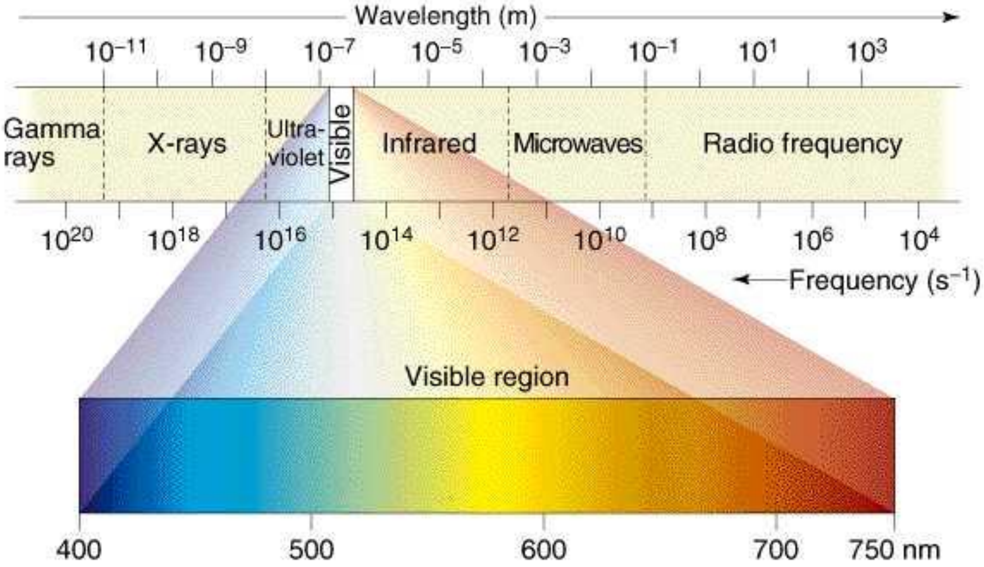
$$m = 9.1\,x\,10^{-31}\ kg \quad v = 10^{6}\ ms^{-1}$$

$$\therefore\ p = mv = 9.1\,x\,10^{-25}\ kg\ m\ s^{-1}$$

$$\therefore\ \lambda = \frac{h}{p}$$

$$= \frac{6.63\times10^{-34}}{9.1\times10^{-25}}$$

$$= 7.3\,x\,10^{-10}\ m$$



This wavelength is comparable in size to an X-ray wavelength.

Calculation of de- Broglie wave:

(i) For the free particle:

The free particle is one which is not bound with any system with any kind of force.

Total energy=Kinetic Energy (E_k)

$$\lambda = \frac{h}{\sqrt{2mE_k}}$$

Calculation of de- Broglie wave:

(ii) For charge particle accelerated with a potential *V*,

If an electron of mass m is accelerated through the potential *V* and velocity *v* then

$$E_k = \frac{1}{2}mv^2 = eV$$

$$v = \sqrt{\frac{2eV}{m}}$$

$$\lambda = \frac{h}{mv} = \frac{h}{m\sqrt{\frac{2eV}{m}}}$$

$$\lambda = \frac{h}{\sqrt{2meV}}$$

Calculation of de- Broglie wave:

Calculate the wavelength associated with an electron subjected to a potential difference of 1.25kV.

Solution,

$$\lambda = \frac{h}{mv} = \frac{h}{\sqrt{2mE_k}} = \frac{6.62\times10^{-34}}{\sqrt{2\times9.1\times10^{-31}\times1.6\times10^{-19}V}}$$

$$\frac{12.28}{\sqrt{V}} = \frac{12.28}{\sqrt{1.25\times10^{3}}} = 0.4\times10^{-10}m = 0.4A^{o}$$

Calculate the de Broglie wavelength associated with a proton moving with a velocity equal to $(1/20)^{th}$ of the velocity of light.

Solution,

$$\lambda = \frac{h}{p} = \frac{h}{mv} = \frac{6.62\times10^{-34}}{1.67\times10^{-27}\times1.5\times10^{7}} = 2.63\times10^{-14}m$$

Calculation of de- Broglie wave:

The de Broglie wave velocity v_ϕ is given by

$$v_\phi = \text{Wavelength} \times \text{frequency} = \lambda f = \left(\frac{h}{p}\right)\left(\frac{E}{h}\right) = \left(\frac{h}{mv}\right) \times \left(\frac{mv^2}{2h}\right) = \frac{v}{2}$$

$$\boxed{v_\phi = \frac{v}{2}}$$

Without includes rest energy:

Einstein's mass- energy relation $E = mc^2$

$$v_\phi = \text{Wavelength} \times \text{frequency} = \lambda f = \left(\frac{h}{p}\right)\left(\frac{E}{h}\right) = \left(\frac{h}{mv}\right) \times \left(\frac{mc^2}{h}\right) = \frac{c^2}{v}$$

$$\boxed{v_\phi = \frac{c^2}{v}}$$

According to Einstein's theory of relativity, the speed of material particle (v) is always less than the speed of light (c). But in above equation the de Broglie wave velocity must be greater than c. This is unexpected result.

Wave function and its Physical interpretation

The wave function ψ can describe both wave and particle nature in quantum mechanics

$$\psi(x,t) = Ae^{-\frac{i}{\hbar}(Et-px)}$$

\# In the beginning the wave function ψ is an ***auxiliary mathematical quantity*** employed to facilitate computations relative to experimental results.

\# Schroedinger said that physical interpretation of wave function ψ is *charge density*. i.e. $|\psi|^2$ is a measure of ***charge density***.

\# If ψ is the amplitude of matter wave at any point in space, then the number of material particle per unit volume is particle density. i.e. *particle density* is proportional to ψ^2.
(In electromagnetic wave energy density is equal to square of amplitude of wave). Thus square of absolute value of ψ i.e. $|\psi|^2$ is a measure of ***particle density***.

Physical interpretation of the wave function:

\# Max-Born suggested that $|\psi|^2 = \psi^*\psi$ represents the ***probability density*** of particle .The probability of finding the particle in a volume element $d\tau = dxdydz$ about any point r at time t is expressed as $p(r)d\tau = |\psi(r,t)|^2 d\tau$

$$\psi(x,t) = Ae^{-\frac{i}{\hbar}(Et-px)}$$

$$\psi^*(x,t) = Ae^{\frac{i}{\hbar}(Et-px)}$$

$$\psi(x,t)\psi^*(x,t) = Ae^{-\frac{i}{\hbar}(Et-px)}Ae^{\frac{i}{\hbar}(Et-px)} = A^2$$

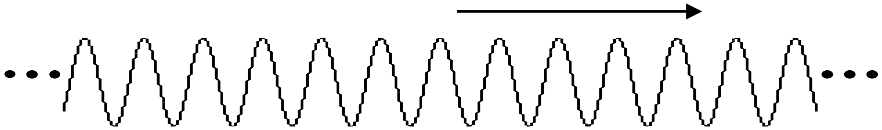
Wave Packet:

Consider a plane wave of single frequency (v) and wavelength (λ) of a wave function

$$\psi_k(x,t) = A \sin 2\pi\left(\frac{x}{\lambda} - vt\right) = A \sin(kx - wt)$$

The velocity of a point in the wave which has a certain fixed phase, therefore the phase velocity.

$$u = \frac{w}{k} = \lambda f$$

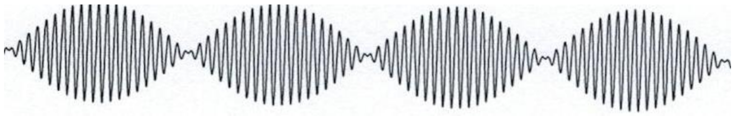


Let two waves having somewhat different frequencies and wavelengths, we get

$$\psi(x,t) = \psi_{k+\Delta k}(x,t) + \psi_k(x,t) = A \sin[(k+\Delta k)x - (w+\Delta w)t] + A\sin(kx - wt)$$

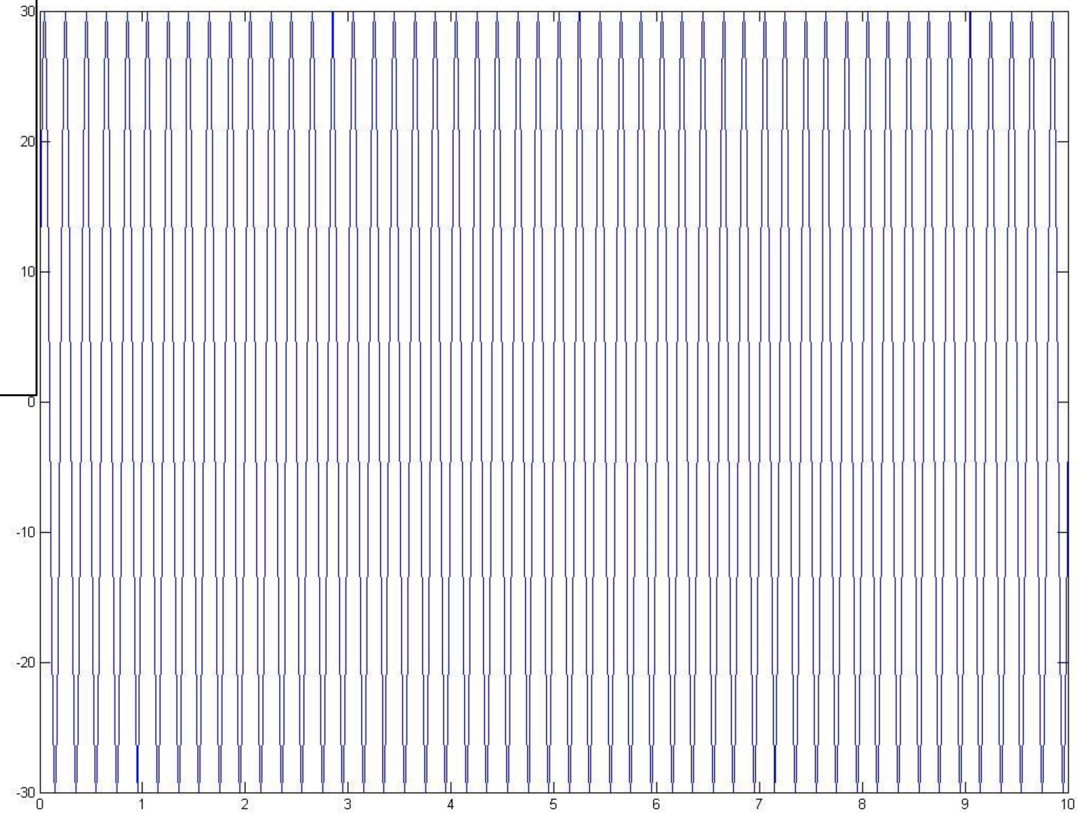
$$= 2Sin\left(\frac{[(k+\Delta k)x - (w+\Delta w)t + (kx - wt)]}{2}\right)\cos\left(\frac{[(k+\Delta k)x - (w+\Delta w)t - (kx - wt)]}{2}\right)$$

$$= \left[2A\cos\left(\frac{\Delta k}{2}x - \frac{\Delta w}{2}t\right)\right]\sin(k'x - w't)$$



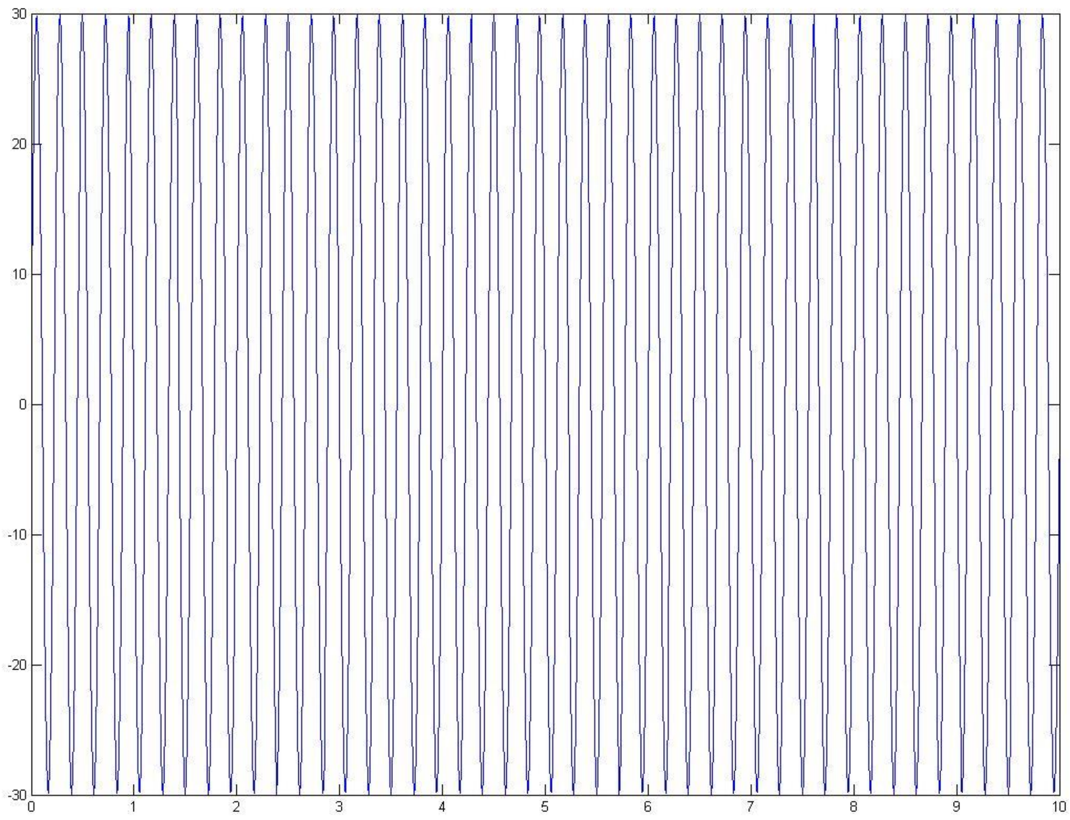
Wave Packet:

```
clear
clc
t=0:0.01:10;
y=30*sin(2*pi*5*t);
plot(t,y)
```



Wave Packet:

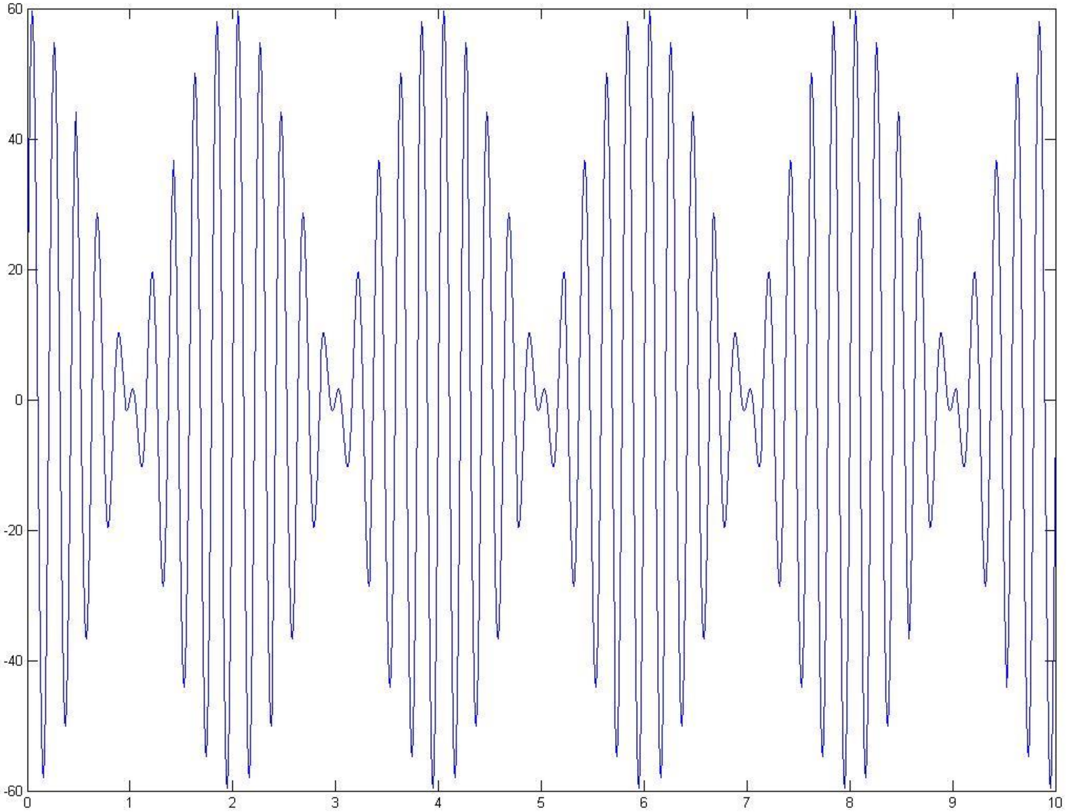
x=30*sin(2*pi*4.5*t);
plot(t,x)



Wave Packet:

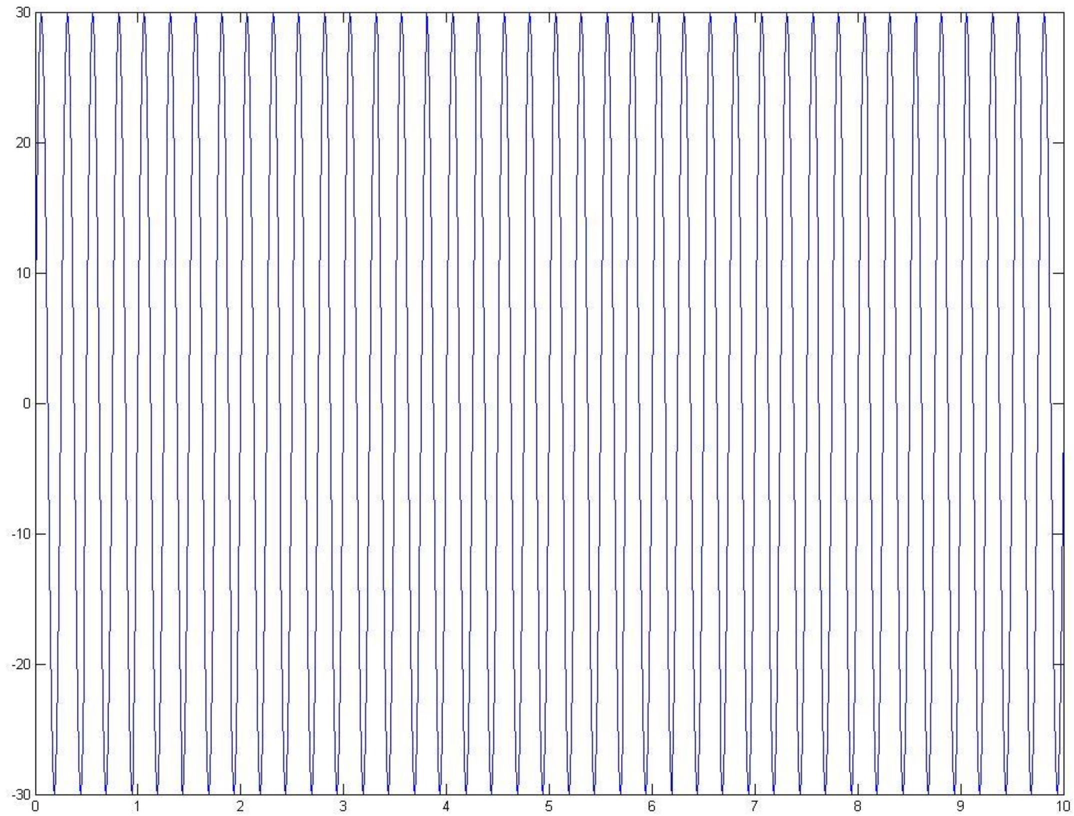
z=x+y

plot(t,z)



Wave Packet:

m=30*sin(2*pi*4*t);
plot(t,m)

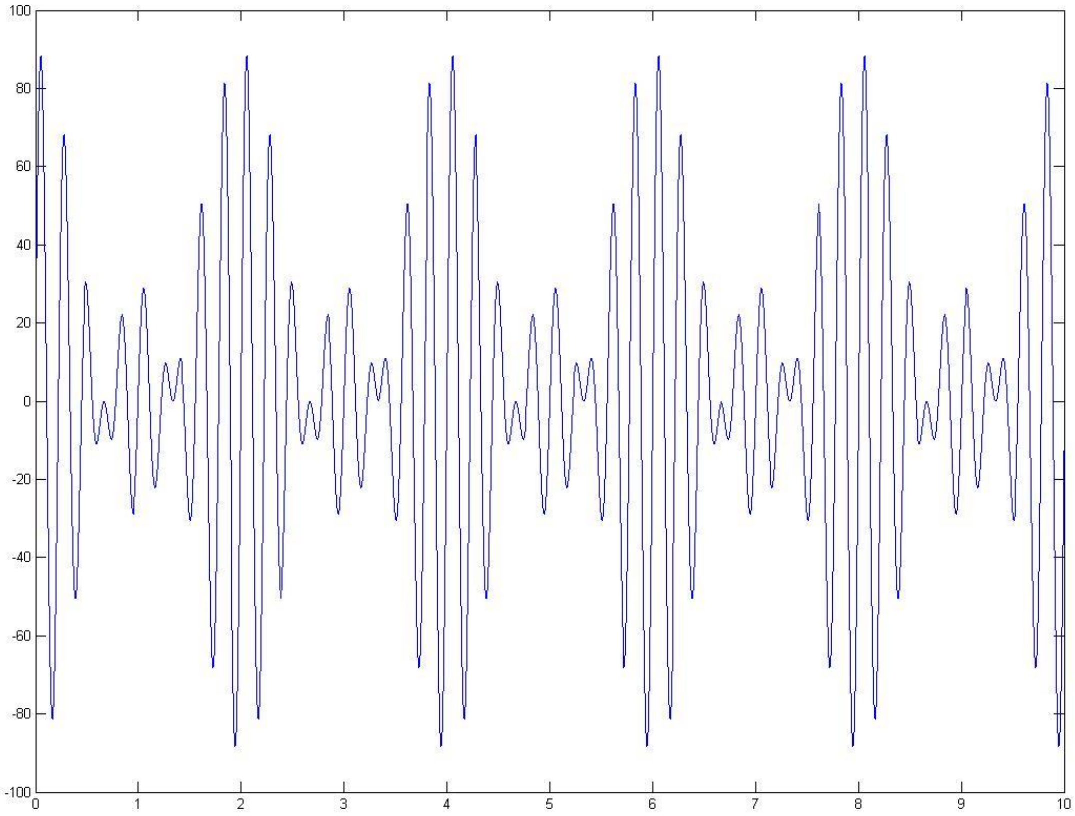


Wave Packet:

z=x+y

n=z+m

plot(t,n)



Wave Packet:

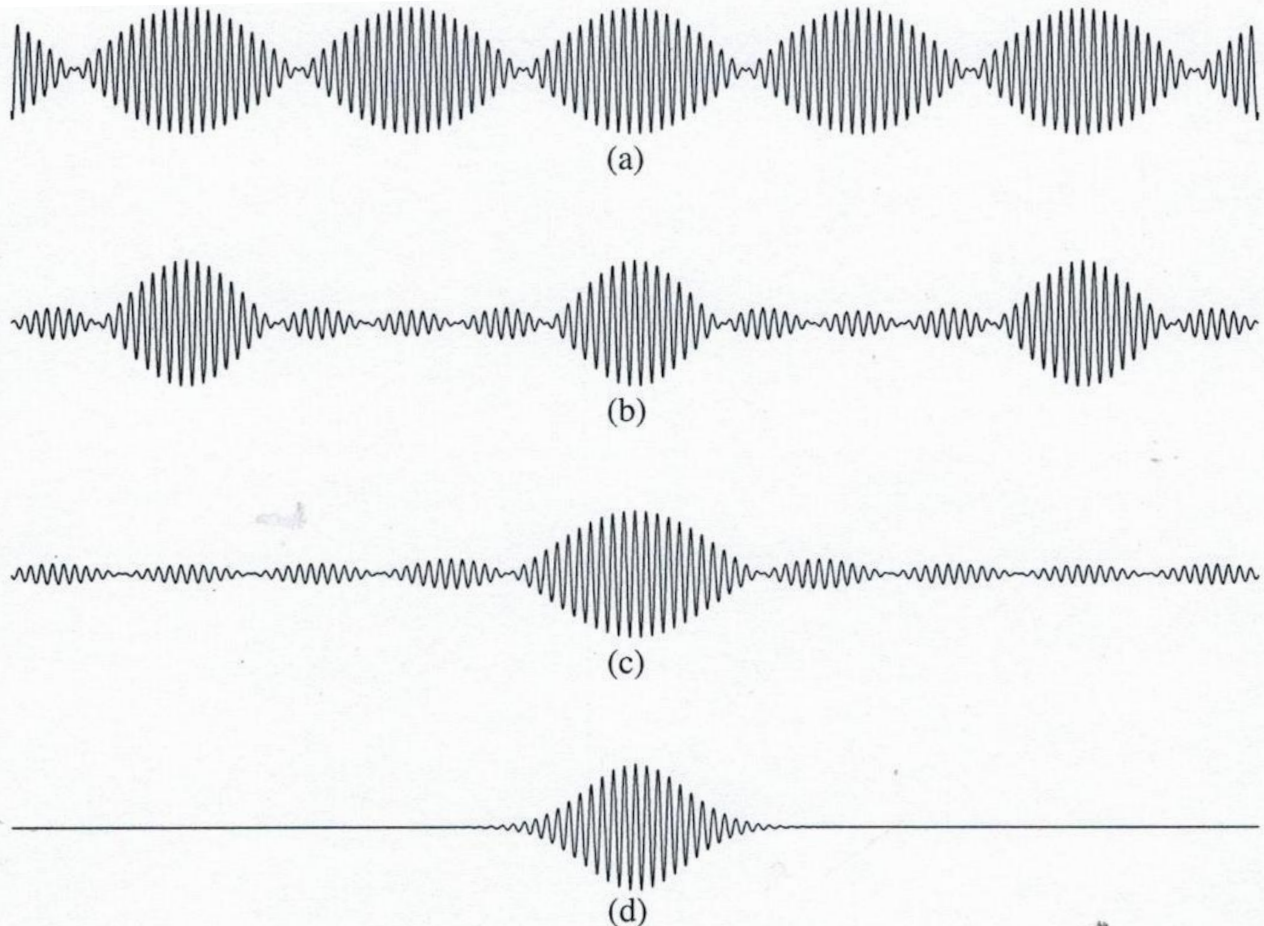
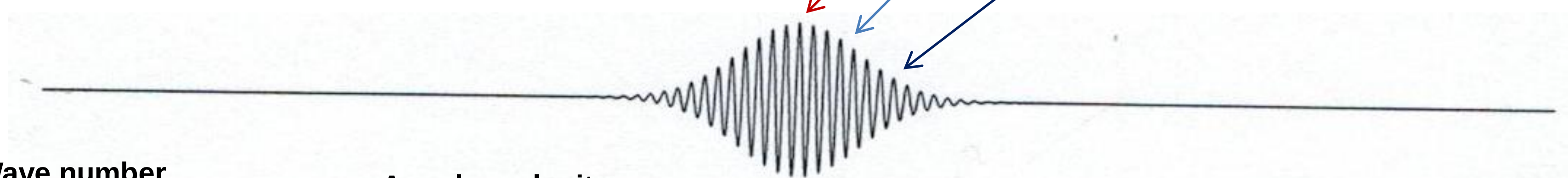


Figure 3.2: (a) Beat notes produced by adding together two cos waves: $\cos(5x) + \cos(5.25x)$.
(b) Combining five cos waves: $\cos(4.75x) + \cos(4.875x) + \cos(5x) + \cos(5.125x) + \cos(5.25x)$.
(c) Combining seven cos waves: $\cos(4.8125x) + \cos(4.875x) + \cos(4.9375x) + \cos(5x) + \cos(5.0625x) + \cos(5.125x) + \cos(5.1875x)$.
(d) An integral over a continuous range of wave numbers produces a single wave packet.

Wave Packet:

From a continuous coherent *superposition* of infinite number of waves (sine, cosine or plane), we can obtain a single maximum *(wave packet).*

$$\psi(\mathrm{x},\mathrm{t}) = \sum_{\mathrm{j}=0}^{\infty} \mathrm{A}_{\mathrm{j}} \mathrm{e}^{\mathrm{i}(\mathrm{k}_{\mathrm{j}} x - w_j t)} = \int A(k) \mathrm{e}^{\mathrm{i}(\mathrm{k}x - w(k)t)} dk$$



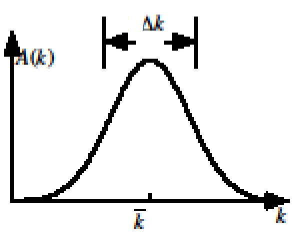
Wave number

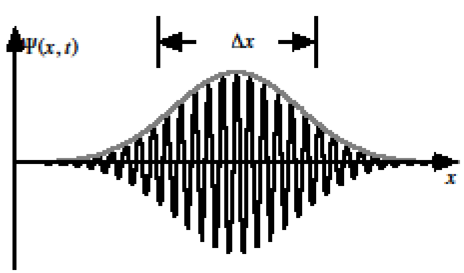
$$k = \frac{2\pi}{\lambda}$$

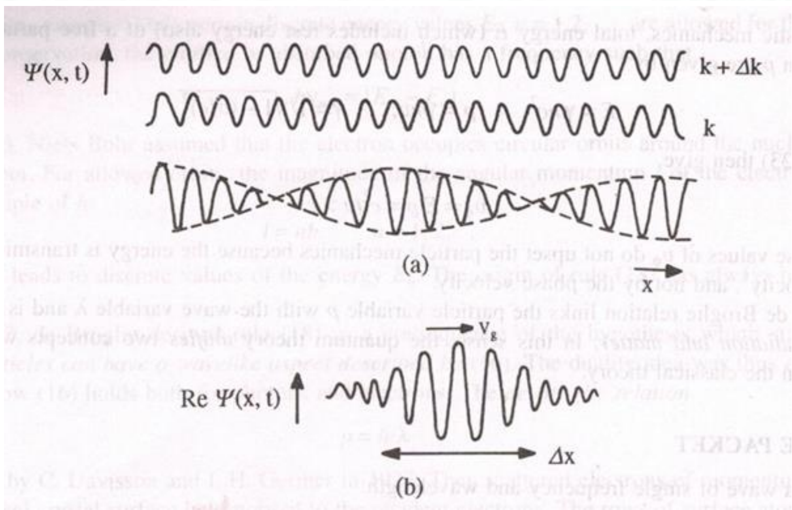
Angular velocity

$$w = 2\pi f$$

Angular Frequency







Relation Between Phase Velocity and Group Velocity

Each component wave propagates with a definite velocity is called phase velocity.

For individual wave, the phase velocity is

$$u = \frac{w}{k} = \lambda f$$

The group wave is formed by superposition of two or more waves. We obtain group wave

$$v = \frac{dw}{dk}$$

The group velocity of wave packet determine the motion of particle.

Wave Packet:

We have, wave vector $$k = \frac{2\pi}{\lambda}$$

$$dk = 2\pi\left(-\frac{1}{\lambda^2}\right)d\lambda$$

We have, angular velocity $$w = 2\pi f$$

$$dw = 2\pi df$$

Group velocity,

$$v = \frac{dw}{dk} = \frac{2\pi df}{2\pi\left(-\frac{1}{\lambda^2}\right)d\lambda} = -\lambda^2\frac{df}{d\lambda} = -\lambda^2\frac{d}{d\lambda}\left(\frac{u}{\lambda}\right)$$

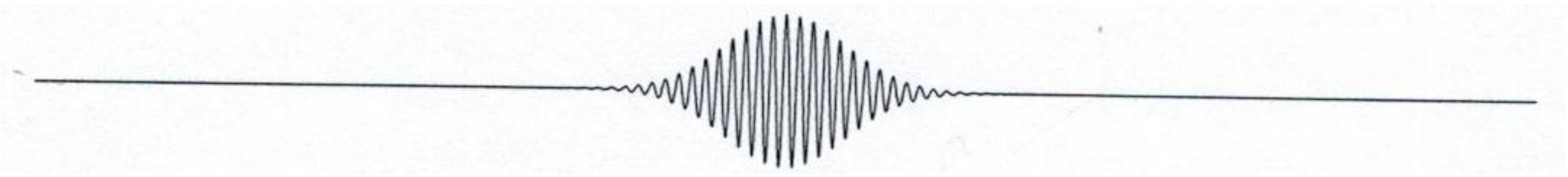
Wave Packet:

Group velocity,

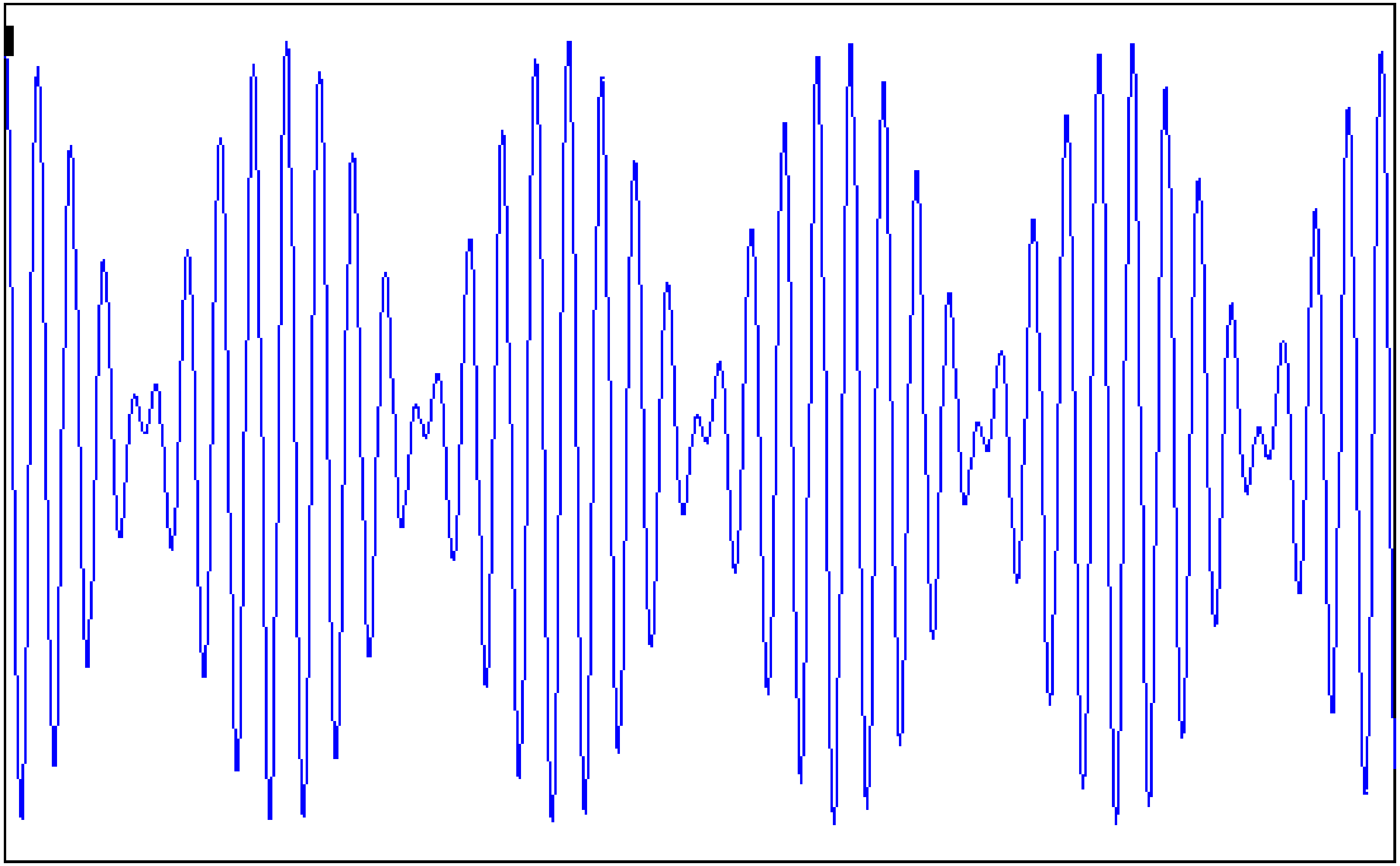
$$v = -\lambda^2 \frac{d}{d\lambda}\left(\frac{u}{\lambda}\right) = -\lambda^2\left[u\frac{d}{d\lambda}\left(\frac{1}{\lambda}\right) + \left(\frac{1}{\lambda}\right)\frac{du}{d\lambda}\right]$$

$$v = -\lambda^2\left[u\left(-\frac{1}{\lambda^2}\right) + \left(\frac{1}{\lambda}\right)\frac{du}{d\lambda}\right]$$

$$v = u - \lambda\frac{du}{d\lambda}$$



Dispersion: phase/group velocity depends on frequency



Black dot moves at phase velocity. Red dot moves at group velocity.

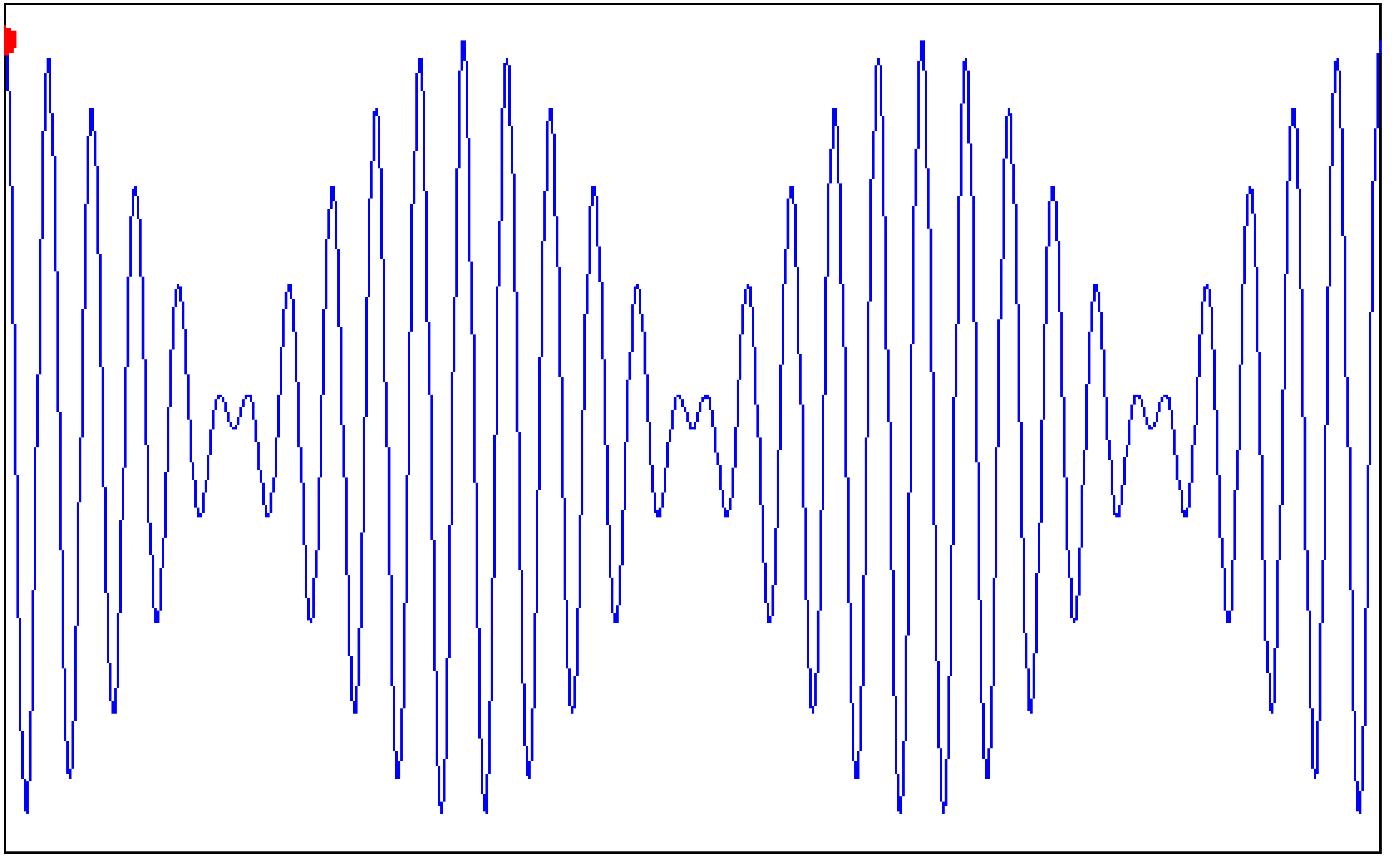
Wave Packet:

$$E = hf = \frac{2\pi hf}{2\pi} = \frac{h}{2\pi} 2\pi f = \hbar w, and, p = \frac{h}{\lambda} = \frac{(\hbar 2\pi)}{\left(\frac{2\pi}{k}\right)} = \hbar k$$

$$v = v_g = \frac{dw}{dk} = \frac{d(\hbar w)}{d(\hbar k)} = \frac{dE}{dp} = \frac{d}{dp}\left(\frac{p^2}{2m}\right) = \frac{2p}{2m} = v_p$$

i.e. velocity of wave packet is identical to velocity of particle.

The Group Velocity



Here, Particle velocity = group velocity